25901153

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

FOURTH SEMESTER

Discipline Specific Elective (DSE) - MG4DSECHE201 – FOOD CHEMISTRY

SCHEME (2024 ADMISSION ONWARDS)

Maximum Marks:70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer questions. Answer any **10** questions Each question carries **2** marks.

1. What are the alkaloids present in pepper? [U]/[CO4]
   Piperine, Chavicine (an isomer of piperine), Piperidine, Piperittine and Piperanine.
   [**4 alkaloids-2 marks, Any 2 alkaloids-1 mark, Any 1 alkaloid-0.5 mark]**

2. What are the different states of water present in foods? [U]/[CO2]

   Free water, bound water and absorbed water.

   [All 3 states-2 marks, Any 2-1 mark, Any 1- 0.5 mark]

3. How does electronic tongue work? [U]/[CO1]

   By using an array of non-specific electrochemical sensors to detect chemical compounds in liquids, similar to human taste buds and analyzing the data with pattern recognition software.

   [Give full marks to all]

4. Give the name of any two enzymes acting on proteins. [U]/[CO3]

   Pepsin, Trypsin, Chymotrypsin, Bromelain & Papain, Carboxypeptidase, Serrapeptase

   [Any 2 enzymes-2 marks, Any 1 -1 mark]

5. What is rendering? [K]/[CO2]

   A thermal process that uses heat to separate pure fat from animal tissues.In this,the fatty tissue is chopped and heated in water,when the fat melts and is released to float above the water.It is further purified by decantation followed by straining.

   [Correct definition-2 marks]

6. Write two important functions and two deficiency disorders of vitamin A. [K]/[CO3]

   - Functions: - It affects various metabolic processes in the body,it has profound effect on epithelial structures, it influences growth in animals and it increases resistance to diseases
   - Deficiency disorders: -Night blindness or nyctalopia (inability to see in dim light or at night), xerophthalmia(hardening of cornea), keratomalacia(softening of the cornea), phrynoderma (hard and horny skin),stunted growth.

   [Any 2 functions and 2 disorders-2 marks,Functions only/disorders only-1 mark.Any 1 function or disorder-1 mark]

7. Why are powdered spices more susceptible to adulteration than whole spices? [An]/[CO4]

Powdered spices are more susceptible to adulteration than whole spices because:

- Loss of identity – In powdered form, the original shape, size and structure of the spices are destroyed, making it difficult to visually identify purity.
- Easy mixing of adulterants- Cheap substances like starch, husk or colored powders can be easily mixed and go unnoticed in the fine powder

Thus, adulteration is harder to detect in powdered spices compared to whole spices

[Correct explanation- 2 marks]

8. What is shelf life of food? [U]/[CO1]

Shelf life of a food indicates how long food retains quality and safety,starting from production and ending based on storage conditions.

[Any relevant explanation – 2 marks]

9. What is the role of yeast in wine production? [K]/[CO4]

Yeast is used for the alcoholic fermentation of grapes or grape juice in wine production.Other roles include flavor and aroma development,controlling sensory components,texture modification,ensuring fermentation success.

[Correct explanation-2 marks.]

10. Name the enzyme and the catalyst required for enzymatic browning. [K]/[CO2]
    - Enzyme – Polypheno oxidase(PPO). Also called tyrosinase orCatechol oxidase.
    - Catalyst- Copper

[Correct answer – 2 marks]

11. Mention one contribution of food chemists in improving nutrition.

Contributions include fortifying foods with essential nutrients,developing healthier substitutes,increasing nutrient bioavailability,extending shelf life,removing contaminants to combat malnutrition and prevent diseases.

[**Any 1 contribution or reasonable points-2 marks]** [A]/[CO3]

12. What is the function of Ascorbic acid as flour treatment agent? [U]/[CO1]

Ascorbic acid or Vitamin C is a common oxidizing agent that strengthens the gluten network (elastic,stretchy and web like structure formed when flour proteins are mixed with water)and improves dough stability.

[Correct explanation 2 mark.Any 1 function related to question- 1 mark]

[2x10=20]

Part B

Short Answer Questions Answer any **6** questions

Each question carries **5** marks

10. Write a short note on artificial food preservatives. [U]/[CO1]

Artificial or chemical food preservatives are chemical substances added to food to prevent spoilage caused by microorganisms such as bacteria, fungi, and yeast.They help in extending shelf life, maintaining flavour, colour, and quality of food.These preservatives are added in limited quantities as permitted by food safety laws, and their presence must be declared on food labels.

Types of Artificial Preservatives

(a) Organic Preservatives-Derived from organic compounds

Common examples:Benzoic acid and sodium benzoate – used in fruit juices, soft drinks;Esters of p-hydroxy benzoic acid(Parabens) – used in pickles,softdrinks,syrups,beer and baked foods;Sodium or potassium salts of sorbic acid – used in processed cheese,pickles,bahed foods,beverages,fish and meat products;Epoxides- such as ethylene oxide and propylene oxide- used in spices,dried fruits and nuts.

(b) Inorganic Preservatives- Simple chemical substances

Examples:Sodium and Potassium metabisulphites(Sulphur dioxide (SO_2)) – used in dried fruits;Sodium and Potassium Nitrates and nitrites – used in meat preservation

Functions include: Inhibit microbial growth,Prevent fermentation and spoilage,Increase storage life,Maintain food quality

Advantages are :Increase food availability and storage time,Reduce food wastage,Enable transport of food over long distances,

Disadvantages are :Excess use may cause health issues,Some may cause allergic reactions,

Strict regulation is required.

[Definition+Types with 1 example each+Any 2 functions – 5 marks, Definition+types with examples-4 marks, Definition+types-3 marks, Definition/Types-2 marks, Function/advantages/Disadvantages only-1 mark]

11. Explain the role of oligosaccharides in food flavor and stability. [U]/[CO2]

Oligosaccharides are carbohydrates that yield 2–9 monosaccharide units on hydrolysis. They include disaccharides (e.g., sucrose) and trisaccharides (e.g., raffinose), tetrasaccharides (eg.lychnose). Their role in food systems is important for both flavor and stability.

(a) Role in Flavor- Sweetness contribution(Disaccharides like sucrose impart characteristic sweetness and improve palatability of foods),Flavor enhancement(Oligosaccharides can enhance natural flavors by balancing acidity and bitterness), Maillard reaction(Some oligosaccharides especially reducing sugars participate in Maillard browning with amino acids, producing desirable flavors, aromas, and color in baked and roasted foods),Fermentation substrates(They act as substrates for microorganisms, leading to production of organic acids, alcohols, and flavor compounds in fermented foods).

(b) Role in Stability- Moisture retention (Oligosaccharides help in water binding, maintaining softness and preventing drying of food products), Texture improvement (They contribute to viscosity and mouthfeel, improving the consistency of foods like syrups and dairy products), Shelf-life extension (By controlling water activity, they inhibit microbial growth and increase storage stability),Thermal stability (Certain oligosaccharides resist breakdown at moderate temperatures, helping maintain structure during processing).

[Definition +any 2 roles each for flavor and stability-5 marks,Roles in flavor only/stability only-3 marks,Definition/Classification/Examples only-1 mark]

12. Discuss the functional properties of minerals in foods. [K]/[CO3]

General functions:

- Minerals form the structural components of bones,teeth,soft tissue,bloodand muscles.eg:Ca,P,Mg in bones
- Regulate activity of nerves with regard to stimuli and contraction of muscles.eg:Cl
- Maintain acid-base balance of body fluids.eg:Na,Cl
- Maintain water balance by means of osmotic pressure and permeability of cell membranes.eg:Na,Cl
- They are constituents of vitamins.eg S in Vit B1,Co in Vit B12
- Form part of molecules of hormones and enzymes.eg:I in thyroxine,Zn in insulin.
- Activate enzymes.eg:Ca activates enzyme lipase
- Regular cellular oxidation.eg: Fe,Mn
- They are necessary for clotting og blood.eg:Ca.

[Any 5 functions-5 marks,Any 4 fns-4marks,Any 3 fns-3 marks,Any 2 fns-2 marks,Any 1 fn-1 mak.Examples only-1 mark]

13. Discuss the nutritional functions of proteins in the human body. [U]/[CO2]

Proteins perform a variety o functions in the human body and are vital for growth, maintenance, and overall health.Three main functions are structural function,regulatory function and energy function.The different clsses of proteins on the basis of their biological functions are as follows:

- Structural Function- Proteins such as keratin and collagen provide mechanical support and structure to tissues like skin, hair, nails, and connective tissues.
- Contractile and Movement Function-Proteins like actin and myosin are responsible for muscle contraction and movement, including heartbeat and body motion.
- Regulatory Function-Some proteins act as hormones (e.g., insulin and glucagon) and regulate important physiological processes like metabolism and blood glucose levels.
- Enzymatic Function -Enzymes are protein in nature and act as biological catalysts, speeding up biochemical reactions in the body (e.g., pepsin and trypsin in digestion).
- Storage and Protective Function-Storage proteins (like ferritin) store essential nutrients such as iron, while protective proteins (antibodies or immunoglobulins) help defend the body against infections.

- Energy Source (Supplementary)-When required, proteins can be used as a source of energy, providing about 4 kcal per gram.

- Transportation Function-Some proteins like transferrin is used for the transporting ferric ions in the plasma.

[Any 5 functions with explanation-5 marks,Any 4 fns-4 marks,Any3 fns-3 marks and so on.]

14. Explain how acidity helps in food preservation. [U]/[CO1]

Acidity plays an important role in preserving food by controlling microbial growth and maintaining food quality. Food acids are compounds added to food products as acidulants or found naturally in foods to add a sharp flavour,preserve the food and act as antioxidants.Acidity favours food preservation in the following ways:

- **Inhibits Microbial Growth**: Low pH created by acids prevents the growth of bacteria, yeasts, and molds. Many harmful microorganisms cannot survive in acidic conditions.
- **Controls pH of Food**:Food acids regulate the pH level, creating an unfavorable environment for spoilage organisms and helping extend shelf life.
- **Acts as Preservative**: Acids like citric acid, acetic acid, and lactic acid are widely used as preservatives in foods such as pickles, jams, and beverages.
- **Enhances Antimicrobial Activity**: Acetic acid (in vinegar) has strong antimicrobial properties and is commonly used in pickled and preserved foods.
- **Maintains Quality and Stability**: Acidity helps in retaining flavour, texture, and color of food while preventing chemical spoilage.
- **Provides Antioxidant Effect**:Some acids like ascorbic acid (Vitamin C) act as antioxidants, preventing oxidation and spoilage.

Examples are:Citric acid(in lemons ,oranges,grapes),malic acid(in apple,plum,cherry),tartaric acid(grapes,pineapple),acetic acid(in vinegar),ascorbic acid(citrus fruits,berries),lactic acid(diary products)soft drinks),carbonic acid(carbonated drinks),phosphoric acid(soft drinks).

[Any 5 fns wih any 2 examples-5 marks,Any 4 fns-4 marks,Any 3 fns-3 marks and so on.Definition/examples-1 mark]

15. Explain the Maillard reaction and its impact on protein quality. [A]/[CO2]

The Maillard reaction is a non-enzymatic browning reaction that occurs between amino acids (from proteins) and reducing sugars when food is heated. It is also called the carbonylamine reaction or protein sugar reaction and was named after the French chemist Louis-Camille Maillard. This reaction leads to the formation of brown colour, flavour, and aroma compounds in foods.Rn occurs during heating or prolonged storage of foods.It involves reaction between free amino groups of proteins and carbonyl groups of sugars.It produces desirable flavours in foods like bread crust, roasted coffee, bakery products, and meat.

Impact on Protein Quality:

- Loss of essential amino acids – especially lysine, as it reacts with sugars and becomes unavailable.
- Reduction in protein digestibility due to formation of complex compounds.
- Decreased nutritional value of proteins in foods like dried milk powder during storage.
- Formation of undesirable browning under low pH and low temperature conditions during long storage.
- In some cases, contributes positively by improving taste and aroma, but nutritionally it lowers protein quality.

[Correct explanation with impact-5 marks,Explanation of Rn-3 marks,Impact only-3 marks]

16. Write a short note on vitamin K and vitamin C. [U]/[CO3]

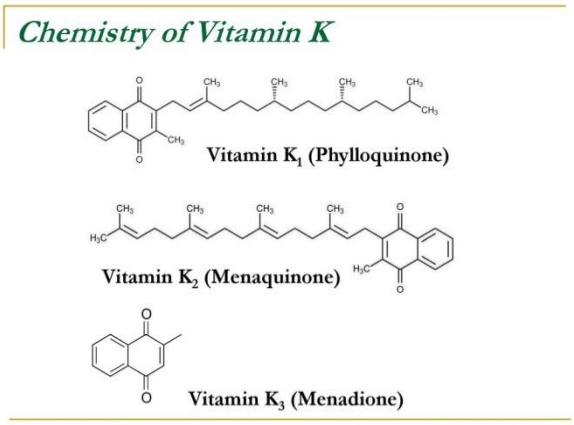
Vitamin K is a fat-soluble vitamin essential for blood coagulation, bone metabolism, and regulating calcium levels in the body. Known as the "clotting vitamin," or “antihaemorrhagic vitamin” it plays a vital role in enabling blood to clot and stopping wounds from bleeding.

Main Forms and Sources:Vitamin K1 (Phylloquinone -Primarily found in green leafy vegetables (spinach, kale, broccoli, lettuce and vegetable oils. This is the main dietary source.),Vitamin K2 (Menaquinone- Produced by bacteria in the human gut and found in fermented foods like Natto and some animal products like eggs, cheese),Vitamin K3 (Menadione-A synthetic form that is not used in human nutrition due to potential toxicity.

Key Functions: Blood Coagulation (Essential for creating proteins (such as prothrombin) that trigger blood clotting factor production in the liver),Bone Health(Aids in the production of osteocalcin, a protein needed to maintain strong bones and increase bone mineral density),Cardiovascular Health(Helps activate matrix proteins MGP that prevent calcium from depositing in arterial walls, thereby protecting against vascular calcification).

Deficiency and Symptoms:Vitamin K deficiency is rare in adults because of its prevalence in the diet and the bacterial synthesis in the gut. However, it can occur in cases of severe malnutrition, fat malabsorption disorders (like Crohn's or Celiac disease), or long-term antibiotic use.Symptoms include increased bruising, nosebleeds, and

bleeding that does not stop easily.Newborns: Infants have limited Vitamin K storage, making them prone to Vitamin K Deficiency Bleeding (VKDB), which is why newborns often receive a Vitamin K shot at birth.

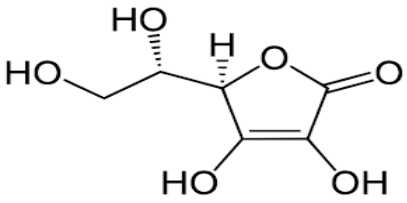


Vitamin C also known as ascorbic acid, is a water-soluble vitamin essential for human health. Because the body cannot synthesize it, it must be consumed daily through diet.

Key Functions:Collagen Synthesis(Crucial for forming collagen, a protein required for healthy skin, cartilage, tendons, ligaments, and blood vessels.),Antioxidant Protection(Acts as a powerful antioxidant that protects cells from damage caused by free radicals.),Immune System Support(Supports immune function and helps white blood cells fight infections),.Iron Absorption(Enhances the absorption of non-heme iron,the iron from plant-based foods),Wound Healing(Essential for repairing tissues and forming scar tissue.)

Sources:Richest Sources: are kakadu plum, camu camu, acerola, rose hips, and guava.Common Sources are Citrus fruits (oranges, lemons), bell peppers, strawberries, kiwi, broccoli, and tomatoes.

Deficiency and Toxicity:A severe lack of vitamin C leads to scurvy, characterized by fatigue,swelling and bleeding of gums, easy bruising, and poor wound healing, loosening of teeth, haemorrhage on the skin and mucous membrane.



[Sources,functions and structure of both vitamins-5 marks,functions and sources-3 marks,functions only-2 marks,structure only-2 marks,Source and name only-1 mark.]

17. Write a brief account of the therapeutic importance of turmeric. [U]/[CO4]

Turmeric (Curcuma longa) is a widely used medicinal plant due to the presence of phenolic compounds called curcuminoids.The three main curcuminoids are curcumin,dimethoxy curcumin and bisdemethoxy curcumin.Curcuminoids are responsible for the yellow colour.Curcumin is the major curcuminoid .Medicinal uses include:

- Anti-inflammatory action-Curcumin helps reduce inflammation by inhibiting inflammatory pathways, making turmeric useful in conditions like arthritis and muscle pain.
- Antioxidant properties-Turmeric neutralizes free radicals and protects cells from oxidative damage, thereby slowing aging and preventing chronic diseases.Turmerin,a protein isolated from turmeric showed a potent antioxidant property.
- Antimicrobial activity-It exhibits antibacterial, antiviral, and antifungal properties, helping in wound healing and preventing infections.
- Digestive health-Turmeric stimulates bile secretion and improves digestion. It is commonly used to relieve indigestion and bloating.Increase the mucin content of gastric juice and reduce irritation in stomach.
- Anticancer potential-Curcumin has been studied for its ability to inhibit the growth of cancer cells and reduce tumor formation.
- Immune system support-Regular consumption enhances immunity and helps the body fight infections more effectively.

- Detoxification of toxic metabolites
- Relieve sore throat,cough,cold etc
- Potent weapon against Alzheimer's disease.
  [Any 5 medicinal uses-5 marks,Any 4 uses-4 marks and so on]

[5x6=30]

Part C

Short Essay Questions

Answer any **2** questions
Each question carries **10** marks

21.Explain the properties of lipids with reference to their physical, chemical, and nutritional properties.

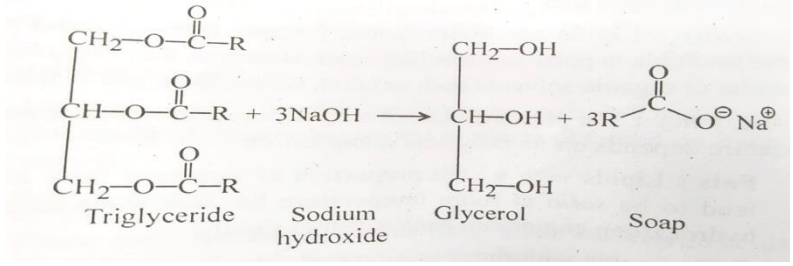
Lipids form a heterogeneous group of organic compounds which are widely distributed in living systems that have similar solubility properties.They are characterized by their insolubility in water but solubility in organic solvents. eg.oils,fats,waxes etc.Properties of lipids can be classified in physical ,chemical and nutritional properties.

a)**Physical Properties** [U]/[C02]

- **Solubility** – Lipids are hydrophobic and are therefore insoluble in polar solvents like water.However they are soluble in nonpolar or organic solvents such as ether ,chloroform and benzene.
- **Consistency**- The physical state of a lipid (liquid or solid) at roomtemperature depends on its fatty acid composition.Lipids with a high proportion of saturated fatty acids(fats) tend to be solid at room temperature because their straight hydrocarbon chains can pack together tightly.Lipids with more unsaturated fatty acids(oils) are typically liquid at room temperature because the double bonds in their hydrocarbon chains cause "kinks, " preventing tight packing.
- **Density** : Lipids are less dense than water, causing them to float.
- **Taste and odour** : Pure lipids are generally colourless, odourles,and tasteless. However, they can absorb flavours and odours from other substances
- **Energy content** : Lipids are energy-rich molecules, providing a high energy yield for life processes.
- **Charge** : Lipid molecules lack an ionic charge, contributing to their hydrophobic (water-repelling) nature.
- **Amphipathic nature** : Some lipids, like phospholipids, are amphipathic, meaning they have both a water-loving (hydrophilic) polar head and a water-fearing (hydrophobic) non-polar tail. This allows them to form cell membranes in an aqueous environment.

b)**<u>Chemical properties</u>**

- **Hydrolysis**: The ester bonds in lipids (triglycerides) can be broken down by reacting with water in a process called hydrolysis. This can be done with either acid or alkali.In acid hydrolysis , triglycerides breaks down into glycerol and fatty acids. In alkaline hydrolysis(saponification) lipid hydrolyse with a strong base (like NaOH or KOH) to form glycerol and fatty acid salts, which are known as soaps.



  - **Hydrogenation**: Unsaturated fatty acids, which contain carbon-carbon double bonds, can react with hydrogen gas in the presence of a catalyst (like nickel).This reaction saturates the fatty acid chains by hydrogenating the double bonds, converting a liquid oil into a solid fat. In practice, the hydrogenation is carried out by passing hydrogen into the heated oil in the presence of powdered nickel as a catalyst. After hydrogenation, the oil is cooled and the catalyst is removed by filtration.The process is frequently used by the vegetable fat or vanaspati ghee manufacturers.



- **Halogenation**:Halogens such as iodine or bromine can be added across the double bonds of unsaturated fatty acids,decolourising the halogen solution.The iodine number or iodine value measures the degree of unsaturation in a lipid.It is defined as the number of grams of iodine that combines with 100g of oil or fat.

c) **Nutritional Properties**

Lipids are essential nutritional components that serve as a concentrated energy source, store energy for later use, and facilitate the absorption of fat-soluble vitamins. They also play vital roles in cell membrane structure, hormone production, insulating and protecting organs, and contributing to the taste, texture, and satiety of food.

- **Energy and Storage**- Lipids provide 9 kilocalories per gram, more than double the calories from carbohydrates or proteins, making them an efficient energy source.(High calorific density).The body stores excess energy as fat serving as a long-term energy reserve for times of need(Energy reserve).

- **Nutrient** Absorption and Transport -Lipids are crucial for the absorption of fat soluble vitamins A, D, E, and K.(Fat soluble vitamins). By combining with fat-soluble nutrients, lipids increase their bio-availability, meaning the body can absorb them more efficiently(Bio availability). Lipids also enhance the absorption of beneficial plant compounds.called phytochemicals, such as lycopene and beta carotene.

- **Cellular and Systemic Functions**- Lipids are vital components of all cell membranes, providing structural support and integrity(Cell structure).They form the myelin sheath that insulates nerve cells, protecting them and enabling quick signal transmission throughout the nervous system(Nervous system). Lipids are necessary for the synthesis of vital hormones, including steroid hormones like estrogen and testosterone, which are involved in reproductive health, stress management, and metabolism.(Hormone production).

- **Protection and Insulation**-Lipids help regulate body temperatuire, providing insulation to help keep the body warm.(Insulation).Adipose tissue provides a physical cushion, protecting vital organs from injury and damage(Cushioning).

- Food Properties-In foods, lipids contribite to desirable aromas, tastes, and textures, meal(Taste and texture).They also promote feelings of fullness or satiety after a meal(Satiety).

[Any 3 physical, chemical and nutritional properties each with explanation-10 marks(minimum of 15 points).Any 2 class of properties with explanation-7 marks.Any 1 class of properties-4 marks]

22.What are essential minerals?Explain nutritional roles of essential minerals in human health? [U]/[CO3]

Essential minerals are inorganic elements required by the human body in small amounts for normal growth, metabolism, and maintenance of health. They are inorganic nutrients required for vital body functions, including bone health, fluid balance, muscle contraction, and enzymatic reactions. They are categorized into macrominerals (needed in large amounts) and trace minerals (needed in small amounts), typically acquired through a balanced diet.They must be obtained through diet because the body cannot synthesize them.

- **Essential Macrominerals** (High Requirement) –**Calcium**(Builds strong bones and teeth, enables blood clotting, and regulates muscle contraction, including the heart. Sources: Dairy products, leafy greens, tofu.)**Magnesium**(Maintains muscle/nerve function, immune system, and energy production. Sources: Almonds, spinach, soybeans.)**Potassium**(Key for fluid balance, nerve signals, and muscle function, reducing blood pressure. Sources: Bananas, potatoes, beans.)**Sodium & Chloride**(Essential for fluid balance and blood pressure regulation.)Phosphorus(Supports bone mineralization, cell membrane formation, and DNA synthesis.)**Sulphur**(Crucial for protein structure and metabolism).
- **Essential Trace Minerals** (Small Requirement) **- Iron**(Essential for producing hemoglobin, which carries oxygen in the blood. Sources: Red meat, lentils, beans.)**Zinc**(Vital for immune function, wound healing, and DNA synthesis. Sources: Meat, shellfish, nuts).**Iodine**(Essential for thyroid hormone synthesis, controlling metabolism.) **Selenium**(Acts as an antioxidant to protect cells and aids thyroid function.), **Fluoride** (Strengthens teeth and bones, preventing cavities.)Manganese & Chromium**(**Involved in glucose metabolism and enzymatic processes). **Copper**(Supports Fe metabolism, connective tissue,andimmunity.)

Nutritional Roles of Essential Minerals in Human Health

- Structural role (bones and teeth)-Minerals like Calcium and Phosphorus are important for the formation and strength of bones and teeth.
- Maintenance of fluid and electrolyte balance-Sodium, Potassium, and Chlorine help regulate osmotic pressure, water balance, and pH of body fluids.
- Nerve transmission and muscle contraction-Calcium, potassium, and sodium are essential for proper nerve impulse transmission and muscle contraction, including heart function.
- Formation of blood components-Iron is required for hemoglobin formation and oxygen transport. Copper helps

in iron metabolism.

- Enzyme activation and metabolic functions-Minerals such as Zinc, Magnesium, and Selenium act as cofactors for enzymes involved in metabolism, DNA synthesis, and antioxidant defense.
- Hormone production and regulation-Iodine is essential for thyroid hormone synthesis, which regulates metabolism and growth.

[Definition+types+examples+any 5 functions-10 marks,Functions only-6 marks,Definition and its types-5 marks.

23.Describe the mechanism of food emulsifiers in stabilizing oil water mixtures. [U]/[CO1]

Food emulsifiers are additives used to help mix ingredients that do not naturally combine, such as oil and water. They help to maintain their texture and prevent the ingredients from separating..They blend immiscible ingredients preventing separation, improving texture, and extending shelf life in products like mayonnaise, ice cream, and baked goods. Examples:Lecithin (Used in chocolate, margarine, and baked goods).Polysorbates (Used in whipped toppings, sauces, and beverages.).DATEM (Diacetyltartaric ester of monoglycerides): Frequently used in bread to strengthen dough. Key Functions- Stabilization,Texture & Mouthfeel,Shelf Life Extension,Aeration etc

Mechanism:

Emulsifiers are amphiphilic molecules, ie,they have two distinct parts:-Hydrophilic Head: A "water-loving" end that is attracted to water and a hydrophobic Tail: An "oil-loving" end that is attracted to fats and oils.When added to a mixture of oil and water, the emulsifiers migrate to the interface (the boundary where oil and water meet). They position themselves so that their tails point into the oil and their heads point into the water.The emulsifier accumulates at the interface between the two liquid.The presence of emulsifiers at the interface significantly lowers the interfacial tension. This lowers the energy required to break down one phase into tiny droplets (dispersed phase) within the other (continuous phase).The emulsifier forms a thin physical layer around the dispersed droplets. This protective film around oil droplets, prevent them from grouping back together (coalescing).Charged heads (especially in ionic emulsifiers) coat the droplets with the same charge, causing them to repel each other.Many stabilizers increase the viscosity of the continuous phase, reducing the movement and collision frequency of droplets.

[Mechanism with minimum 7 points-10 marks,definition+function+types-4 marks, Definition /function /types only-2 marks]

24. (i)Discuss the major components present in ginger

(ii) What are the applications of ginger in daily life and medicine? [U]/[CO4]

(i) Ginger is the rhizome (a modified underground stem) of the Zingiber officinale plant. Its unique characteristics include:

- Active Compounds: It contains volatile oils such as zingiberene, ringerone, gingerol, and shogaol.
- Flavor: It has a mildly pungent taste with lemony or camphory notes.
- Dry Ginger: Known as "chukku," it is more pungent than fresh ginger and is highly valued for adding zest to blander dishes.

(ii) **Culinary Uses:**

- Ginger is a versatile ingredient used in various forms:
- Whole Root: Used in curries, pickles, chutneys, and preserved dried fruits.

Ground Ginger: A common "masala" for dishes like pulao, pongal, and various non-vegetarian preparations.

Beverages: Often added to tea, lime juice, and buttermilk for flavor.

Murabba: The text notes that ginger is used in making Murabba, a sweet fruit preserve cooked in sugar syrup.

Medicinal Benefits

Ginger is widely recognized for its therapeutic properties:

- Antioxidant: It possesses strong antioxidant properties.
- Anti-inflammatory: Reported to reduce inflammation and pain in joints.
- Migraine Relief: It has potential prophylactic (preventative) use in treating migraine headaches.
- Digestive Aid: Often used to treat stomach-related issues (implied by its volatile oil content).

- Alleviating nausea and for motion sickness
- Inhibits growth of colon bacteria that cause flatulence
- Reduces intestinal contraction,neutralize digestive acids and inhibit the vomiting centre in brain
- Detoxification of carcinogens
- Natural cough suppressant

Components+Any 5 applications=5+5 marks,Applications only-5 marksAny 4 applications+components=8

[10x2=20]

18. Describethemechanismoffoodemulsifiersinstabilizingoilwatermixtures.
[U]/[CO1]

19. (i)Discussthemajorcomponentspresentinginger
(ii)Whataretheapplicationsofgingerindailylifeandmedicine? [U]/[CO4]

[10x2=20]